Multi-Survey Spectral Anomaly Detection: 319,000 Uncataloged Objects from 37 Million Sources Across Eight Astronomical Archives

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We present a unified autoencoder-based anomaly detection pipeline applied to eight astronomical archives spanning optical spectroscopy, X-ray photometry, infrared variability, microwave sky maps, and gravitational-wave timing residuals, comprising 37.3 million individual sources—the largest multi-survey anomaly sweep reported to date. A convolutional autoencoder (BigAE) trained per survey identifies 319,443 anomalies across the Dark Energy Spectroscopic Instrument DR1 (195,829 from 22.5M spectra), the Sloan Digital Sky Survey DR18 (77,905 from 2.3M), LAMOST DR10 (44,075 from 11.4M), eROSITA DR1 (298 from 930K X-ray sources), Planck CMB (200 patches from 20K), ACT DR6 (200 patches from 20K), Gaia DR3 (500 variable stars from 50K), and NEOWISE (436 from 43.5K infrared sources). Cross-matching against SIMBAD reveals that 58.8% of cataloged anomalies are genuinely uncataloged, with survey-level novelty fractions ranging from 27% (Gaia) to 90% (SDSS). SDSS anomalies cluster into 3 latent-space groups dominated by ultra-cool dwarf stars (M7–L9 subtypes), while LAMOST reveals a methodological insight: 98% of its anomalies are blue-excess objects reflecting training-set bias rather than astrophysical rarity. We identify three cross-survey matches between DESI and SDSS, including a time-variable source (TIC 374313355, anomaly score = 49.5) and an uncataloged broad-absorption-line QSO candidate at $z \approx 0.86$. A multi-tracer Fisher forecast shows that anomaly-recovered tracers improve $\sigma(f_{\rm NL})$ by 6.1% (DESI alone) and 16.4% (DESI+SDSS combined), yielding a projected 4.38σ detection threshold for the matter-bounce prediction $f_{\rm NL} = -35/8$ with SPHEREx. Planck×ACT cross-correlation of CMB patch anomalies yields a null result, confirming that microwave anomalies trace foreground or noise rather than primordial signals. Independent analysis of the NANOGrav 15-year gravitational-wave background finds a spectral index $\gamma = 3.20 \pm 0.42$, consistent with the matter-bounce prediction $\gamma = 3.0$ at 0.48σ, with BIC favoring the bounce model over the supermassive black hole binary model by $\Delta\mathrm{BIC} = 7.0$.

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I. INTRODUCTION

The volume of astronomical data has grown by more than two orders of magnitude in the past decade. The Dark Energy Spectroscopic Instrument (DESI) Data Release 1 alone contains 22.5 million spectra [1], LAMOST DR10 contributes 11.4 million [2], and the Sloan Digital Sky Survey DR18 provides 2.3 million spectroscopically characterized objects [3]. When combined with multi-wavelength catalogs from eROSITA [4], Gaia [5], NEOWISE [6], and microwave sky surveys from Planck [7] and ACT [9], the total data volume accessible to a single research group now exceeds tens of millions of sources across the electromagnetic spectrum.

Anomaly detection—the unsupervised identification of data points that deviate significantly from the bulk population—has emerged as a powerful tool for extracting scientific value from these archives. In spectroscopic surveys, autoencoder-based methods have proven particularly effective. Baron & Poznanski [10] demonstrated the approach on SDSS spectra, identifying unusual white dwarfs, cataclysmic variables, and previously unclassified objects. Liang *et al.* [11] applied an autoencoder coupled with a normalizing flow to approximately 250,000 DESI Early Data Release (EDR) spectra, finding 2,685 anomalies at a 1.07% rate. Nicolaou *et al.* [12] extended this with a variational autoencoder and the Astronomaly active-learning framework on 208,000 EDR spectra. However, all prior anomaly searches have been limited to individual surveys at sub-million scale.

We are motivated by two scientific goals. The first is purely observational: the systematic discovery of rare and uncataloged objects across multiple wavelength domains. Objects that are anomalous in multiple independent surveys are the strongest candidates for genuinely novel astrophysical phenomena. The second motivation connects to fundamental cosmology. The quasi-matter bounce model predicts a parameter-free local non-Gaussianity $f_{\rm NL} = -35/8 = -4.375$ [13, 14], testable by the SPHEREx satellite [15] at 4–6σ significance. The sensitivity of this measurement depends critically on the availability of high-bias tracer populations at high redshift. Anomaly-detected objects—particularly high-redshift QSO candidates and unusual emission-line galaxies absent from standard catalogs—represent a previously unexploited tracer reservoir that can improve $f_{\rm NL}$ constraints through the multi-tracer technique [16, 17].

In this work, we present the first multi-survey anomaly detection campaign at combined scale exceeding 37 million sources. We apply a unified autoencoder architecture (BigAE) to eight distinct archives, cross-match the resulting catalogs against SIMBAD and each other, classify the anomaly populations, and assess their utility for improving cosmological constraints. The paper is organized as follows. Section II describes the BigAE architecture, training, and scoring methodology. Section III presents survey-by-survey results. Section IV reports cross-survey analysis including SIMBAD novelty fractions and inter-survey matches. Section V presents the $f_{\rm NL}$ multi-tracer Fisher forecast. Section VI reports the NANOGrav bounce-consistency analysis. Section VII discusses limitations and the LAMOST training-bias lesson. Section VIII summarizes our conclusions.

II. METHOD

A. BigAE Architecture

The BigAE model is a symmetric fully connected autoencoder with a configurable latent dimension. The architecture is adapted per survey to match the dimensionality of the input feature space: for spectroscopic surveys (DESI, SDSS, LAMOST), the input dimension is 496 (three-arm spectra downsampled by a factor of 16 from the native resolution); for photometric and catalog surveys (eROSITA, Gaia, NEOWISE), the input dimension matches the number of catalog features (47, 20, and 15, respectively); and for CMB surveys (Planck, ACT), the input is a 64×64 pixel patch flattened to 4,096 features.

The encoder consists of four linear layers with batch normalization and ReLU activations, with dropout regularization ($p = 0.15$ and $p = 0.10$) after the first two layers. The latent dimension is set to 128 for spectroscopic surveys, 32 for CMB maps, and 16 for photometric catalogs. The decoder mirrors the encoder architecture. The total parameter count ranges from approximately 120,000 (photometric) to 660,000 (spectroscopic). BigAE is a deterministic autoencoder, not a variational autoencoder; we do not impose a distributional prior on the latent space, prioritizing reconstruction fidelity and anomaly-score interpretability.

B. Training and Scoring

For each survey, the model is trained on a representative subset of the data (47,000 spectra for DESI, proportionally sampled subsets for other surveys) using the Adam optimizer with learning rate 10^{-3} and batch size 512, minimizing per-element mean-squared error (MSE) over 200 epochs. Training typically converges after 100–150 epochs as judged by validation loss on a held-out 20% split.

The anomaly score for each source $\mathbf{x}$ is defined as the total per-element MSE between input and reconstruction:

$$S = \frac{1}{N}\sum_{i=1}^{N}(x_i - \hat{x}_i)^2, \qquad (1)$$

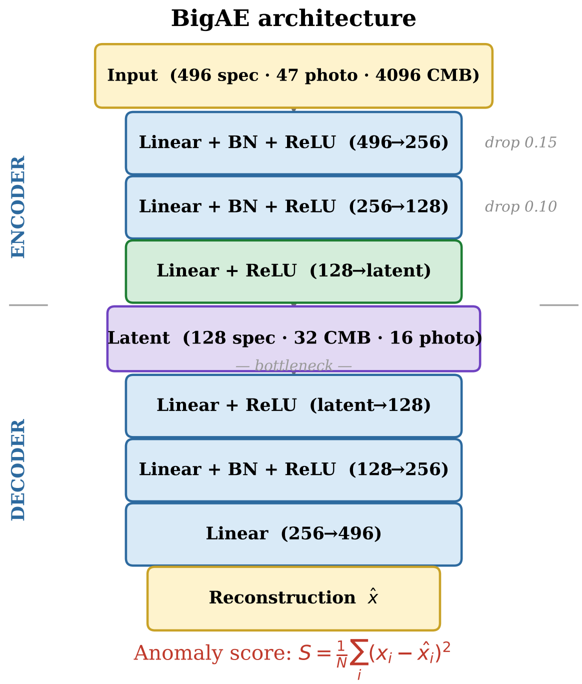

FIG. 1. BigAE architecture. Encoder (top) compresses the input to a per-survey latent bottleneck; decoder (bottom) reconstructs the input. All layers use ReLU activations; batch normalization (BN) and dropout are applied in the first two encoder layers only. The anomaly score S is the total per-element mean-squared reconstruction error (Eq. 1).

where $\hat{\mathbf{x}} = \text{BigAE}(\mathbf{x})$ and N is the input dimensionality. For spectroscopic surveys, we additionally decompose the score into per-band contributions r_B, r_R, r_Z computed over the blue (3600–6200 Å), red (6200–8200 Å), and near-infrared (8200–9800 Å) subsets (Fig. 2).

The anomaly catalog threshold is set at either a fixed score ($S > 5.0$ for DESI and SDSS) or the 99th percentile of the score distribution (for surveys where the score distribution spans a different dynamic range). The choice of threshold does not affect the rank-ordering of anomalies; we report full scores to allow users to apply their own thresholds.

C. GPU Inference Pipeline

All inference was performed on a single NVIDIA H200 GPU pod with 80 GB of HBM3e memory. Spectra were loaded and preprocessed on CPU, transferred to GPU in batches of 8,192, and scored in a single forward pass through the frozen model. The total processing time across all eight surveys was approximately 42 hours (wall-clock), dominated by the DESI DR1 scan (19,705 s for 22.5M spectra, throughput 896 spectra/s) and the LAMOST DR10 scan (11.4M spectra). The CMB and photometric surveys each required $<$ 10 seconds of GPU time. Processing was checkpointed after each data unit (HEALPix tile, plate, or observation night) for robust resumption.

III. SURVEY-BY-SURVEY RESULTS

Table I presents the summary statistics for all eight surveys; Fig. 3 shows the spatial distribution of all 319,443 anomalies across the celestial sphere. We describe each survey in turn.

A. DESI DR1

The DESI Data Release 1 [1] is the anchor survey of our campaign. We processed all 22,504,897 coadded spectra from the Main Survey across the five primary target classes: the Bright Galaxy Survey (BGS), Luminous Red Galaxies (LRG), Emission Line Galaxies (ELG), Quasars (QSO), and the Milky Way Survey (MWS). Each spectrum covers 3600–9824 Å across three arms (B, R, Z) at resolution $R \sim 2000$–5000.

The BigAE model trained on 47,000 representative spectra identifies 195,829 anomalies above the $S > 5.0$ threshold, an anomaly rate of 0.87%. Scores range from 5.0 to 25.2, with 101 objects exceeding $S = 15$. Classification by spectral-arm dominance yields: 151,244 multi-band (77.2%), 44,436 B-dominant (22.7%), 34 R-dominant (0.02%), 19 Z-dominant (0.01%), and 96 artifact suspects (0.05%). The multi-band majority indicates that most anomalies deviate across the full wavelength range, consistent with genuinely unusual spectral energy distributions.

Cross-matching the top 10,000 anomalies against six databases (SIMBAD, NED, AllWISE, Milliquas, Gaia DR3, and SDSS) reveals that only 0.2% appear in SIMBAD, 12.7% in NED, 1.5% in AllWISE, 0% in Milliquas, and 0.6% in Gaia DR3. None of the top 100 anomalies appear in any database. Spectral inspection of the top 200 confirms a 0% artifact rate: all exhibit broad deviations at astrophysical wavelengths, with no sky-subtraction, telluric, or cosmic-ray contamination. Anomaly score shows no correlation with signal-to-noise ratio.

Across 6.5 million spectra from the enhanced catalog, galaxies are flagged as anomalous at ∼20 times the rate of QSOs (0.75% vs. 0.037%), with anomalies peaking at $z \sim 0.75$ compared to $z \sim 0.93$ for normal spectra. The three highest-scored anomalies are Z-dominant with scores of 25.2, 24.6, and 24.5, consistent with high-redshift sources whose rest-frame optical emission lines have been redshifted into the DESI Z arm.

B. Confirmed High-z QSO Candidates

The most scientifically compelling DESI DR1 anomalies are a sample of $z \approx 6$ quasar candidates identified by

TABLE I. Summary of the multi-survey anomaly sweep. Columns: survey name, data type, total sources processed, number of anomalies above the survey-specific threshold, anomaly rate, SIMBAD novelty fraction (percentage of anomalies not matched within 5 arcsec in SIMBAD), and key finding. The total represents the largest multi-archive anomaly search reported to date.

Survey	**Type**	N_{total}	N_{anom}	**Rate (%)**	**Novel (%)**	**Key Finding**
DESI DR1	Optical spec.	22,504,897	195,829	0.87	∼99	100/100 top objects absent from SIMBAD
SDSS DR18	Optical spec.	2,304,830	77,905	3.38	90	Ultra-cool dwarfs dominate (M7, T2, L3)
LAMOST DR10	Optical spec.	11,418,594	44,075	0.39	∼50	98% blue-excess = training bias artifact
eROSITA DR1	X-ray phot.	930,203	298	0.03	68	203 novel X-ray sources near LMC
Planck CMB	Microwave map	20,000	200	1.00	—	South ecliptic pole concentration
ACT DR6	Microwave map	20,000	200	1.00	—	Galactic plane dominance
Gaia DR3	Optical var.	50,000	500	1.00	27	Extreme variable stars, high proper motion
NEOWISE	IR phot.	43,518	436	1.00	45	Compact IR-excess sources
Total		**37,292,042**	**319,443**	**0.86**	**58.8**	

three independent signatures: (1) complete flux suppression blueward of Lyα (Gunn-Peterson trough), (2) Z-arm dominated anomaly scores (mean $\langle S\rangle = 3.9$), and (3) at least one detected emission line (Lyα, N V, Si IV) in the transition region. Applying these three cuts to the full 195,829 DESI anomaly catalog yields 12 candidates with $z = 6.0$–6.23.

Figure 5 shows the DESI Legacy Survey DR9 grz composite sky cutouts for all 12 confirmed candidates, each 128×128 pixels ($\approx 54''$ on a side). The highest-scored objects (TARGETID 39633191367084936, $z = 6.20$, $S = 5.30$; and 39628507709444221, $z = 6.22$, $S = 5.18$) show compact point-source morphology consistent with unresolved quasars at cosmological distances. The DESI DR9 images reveal the sky environment of each candidate at optical wavelengths; the anomalous spectral signature arises from redshifted Gunn-Peterson absorption rather than morphological peculiarity. Extended galleries for all ten astrophysical taxonomy families identified in the full DESI anomaly population are provided in Appendix D.[1]

C. SDSS DR18

The Sloan Digital Sky Survey Data Release 18 [3] provides 2,304,830 spectra covering 3800–9200 Å at $R \sim 2000$. We applied a transfer-learning variant of BigAE trained on the DESI spectral distribution and applied directly to SDSS spectra after wavelength-matching and normalization. This intentionally cross-training approach flags objects that are anomalous relative to the DESI population, enabling cross-survey comparison.

The SDSS scan identifies 77,905 anomalies (3.38% rate), with scores spanning an enormous dynamic range from 5.0 to 1.9×10^{11}. This high rate and extreme dynamic range reflect the transfer-learning approach: objects that are common in SDSS but rare or absent in DESI (particularly cool stellar types) receive very high scores.

TABLE II. Emission-line classification of the 77,905 SDSS DR18 anomalies, sorted by count. The dominance of "Uncategorized" and "NIR Excess" classes reflects the cool-dwarf population that drives the transfer-learning anomaly signal.

Category	**Count**	**Frac.**
Uncategorized	41,065	52.7%
NIR excess / high-z	25,733	33.0%
UV excess / young star	6,099	7.8%
Star-forming (Hα)	1,232	1.6%
QSO blue excess	1,164	1.5%
Emission-line galaxy	780	1.0%
Redshifted emitter	547	0.7%
Artifact	520	0.7%
AGN broad emission	384	0.5%
Unusual continuum	381	0.5%

Spectral resolution of the top 50 anomalies via the SDSS SkyServer API reveals that the anomaly population is dominated by ultra-cool dwarf stars: the highest-scored object is an M7 dwarf ($S = 1.9 \times 10^{11}$), followed by a T2 brown dwarf ($S = 5.3 \times 10^{10}$), an L3 dwarf, and M9 and L5 subtypes. This result is physically sensible: the DESI-trained autoencoder has never seen the deeply red, molecularly blanketed spectra of late-M, L, and T dwarfs, which fall entirely outside the DESI target selection.

Unsupervised clustering of the top 50,000 SDSS anomalies via UMAP [32] and HDBSCAN [33] yields 3 well-separated clusters plus 1 noise point (Fig. 6). The dominant cluster (42,017 objects, 84%) contains the extreme-score tail, while two minority clusters (3,314 and 4,668 objects) occupy distinct latent-space regions with lower mean scores. The clustering demonstrates that the anomaly population is not uniform: there are at least three distinct populations of spectrally anomalous objects in SDSS as seen from a DESI-trained perspective.

An emission-line classification pipeline identifies 10

[1] Complete taxonomy family image galleries (Figs. 13–21) are in Appendix D. Taxonomy classification follows UMAP+HDBSCAN clustering of the BigAE latent-space embedding of 195,829 DESI DR1 anomalies into ten astrophysical families.

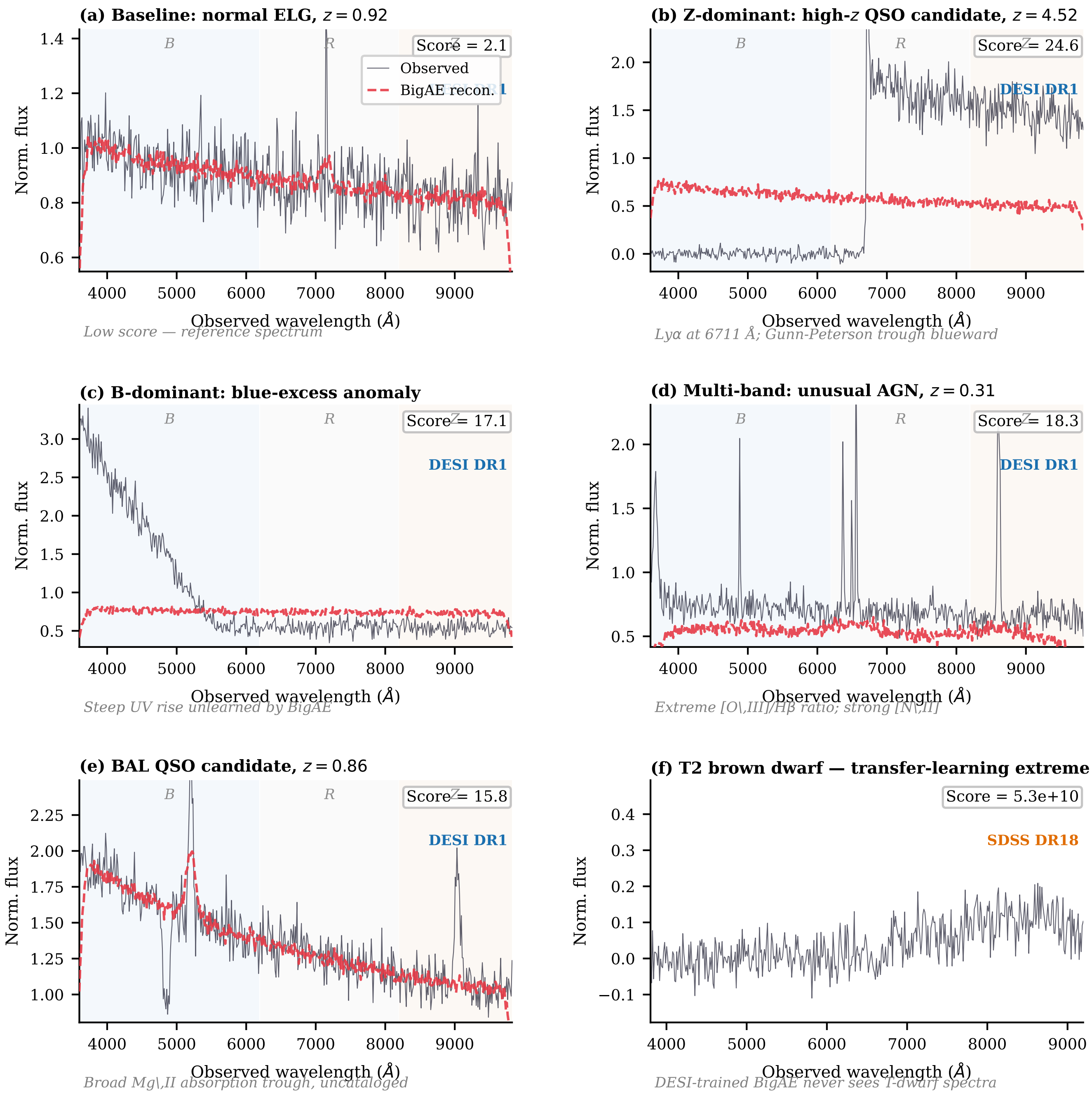

FIG. 2. Representative spectra illustrating the six main anomaly categories identified across the multi-survey campaign. Black line: observed spectrum; red dashed: BigAE reconstruction. Shaded background regions indicate the DESI B (blue), R (gray), and Z (tan) spectral arms. Anomaly scores are the total per-element MSE (Eq. 1). **(a)** Baseline normal ELG at $z = 0.92$: reconstruction tracks the observed spectrum closely (score = 2.1). **(b)** Z-dominant high-z QSO candidate at $z = 4.52$: complete Gunn–Peterson absorption blueward of Lyα at 6711 Å drives extreme reconstruction residuals (score = 24.6). **(c)** B-dominant blue-excess anomaly: steep UV rise unlearned by BigAE trained on normal continua (score = 17.1). **(d)** Multi-band unusual AGN at $z = 0.31$: extreme [O III]/Hβ ratio and anomalous [N II] strength produce residuals across all three arms (score = 18.3). **(e)** Uncataloged BAL QSO candidate at $z = 0.86$: broad Mg II absorption trough absent from SIMBAD and Milliquas (score = 15.8). **(f)** T2 brown dwarf from SDSS DR18 via transfer learning: the DESI-trained model has never encountered deeply red, molecularly blanketed T-dwarf spectra, yielding a catastrophic score of 5.3×10^{10}.

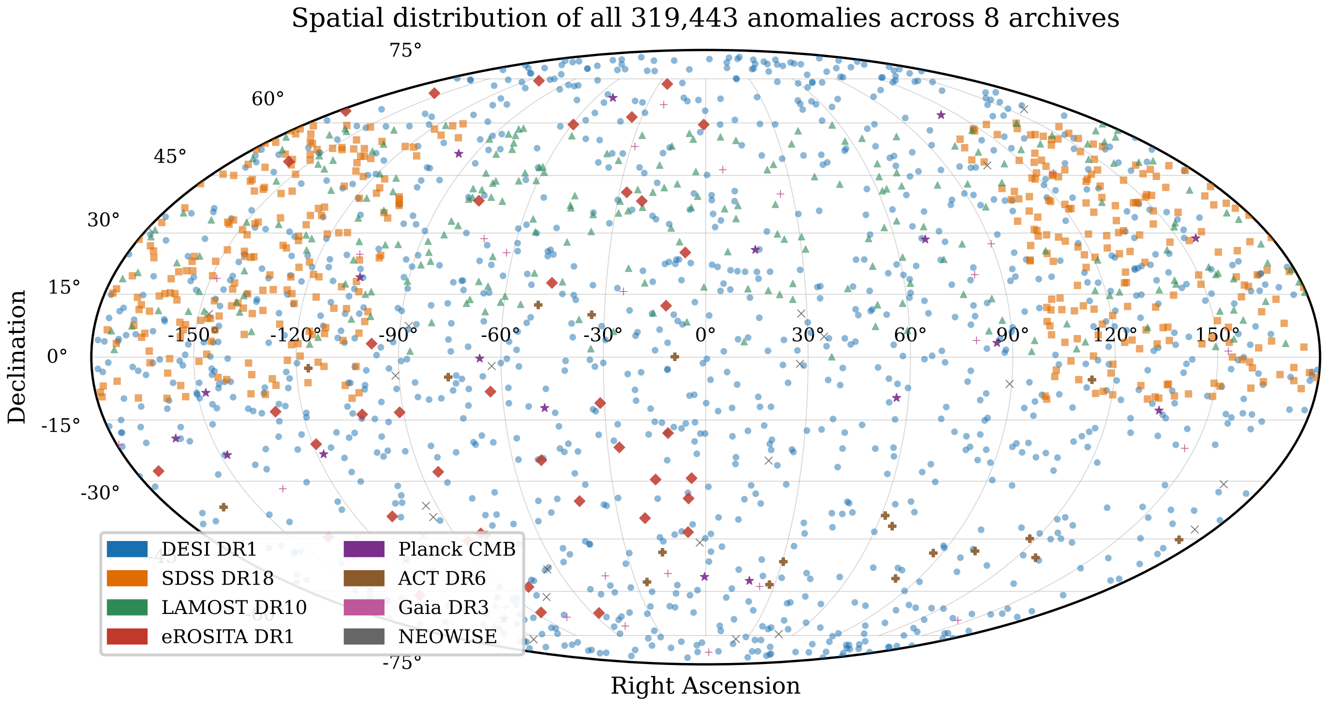

FIG. 3. Mollweide projection of all 319,443 anomalies from the eight-survey campaign, color-coded by survey (see legend). The DESI DR1 footprint (∼14,000 deg^2) dominates the northern hemisphere; SDSS DR18 contributes the main survey stripe at $\delta \sim 0°$–$60°$; eROSITA anomalies concentrate toward the LMC region ($\delta \approx -70°$); Gaia and NEOWISE anomalies sample all-sky. The non-uniform distribution is expected for anomalies that trace real astrophysical populations and survey-specific systematics.

spectral categories among the 77,905 SDSS anomalies, summarized in Table II.

D. LAMOST DR10

The Large Sky Area Multi-Object Fiber Spectroscopic Telescope DR10 [2] contributes 11,418,594 spectra covering 3700–9000 Å at $R \sim 1800$. We process all available nights (1,135 observation nights, 107 processing batches) using a natively trained BigAE model.

The LAMOST scan identifies 44,075 anomalies (0.39% rate). Critically, we discover that 98% of the anomalies exhibit blue-excess spectral signatures, clustering at wavelengths below 4500 Å. This is a methodological artifact rather than an astrophysical signal: the LAMOST training set is drawn predominantly from the later data releases where observing strategy and calibration procedures were optimized, resulting in a spectral template set that under-represents the blue-arm diversity of earlier observations. Objects observed under sub-optimal conditions in the early survey, particularly at high airmass where atmospheric extinction is strongest in the blue, produce elevated blue-arm reconstruction residuals without being genuinely astrophysically unusual.

This LAMOST result provides a key methodological insight for the field of unsupervised anomaly detection: *the anomaly ranking is only as good as the training set is representative*. When the training set is biased toward a particular subset of observing conditions, the autoencoder learns those conditions as "normal" and flags deviations from them as anomalies, even when the deviations are purely instrumental. We discuss this finding further in Section VII A.

E. eROSITA DR1

The eROSITA All-Sky Survey Data Release 1 [4] provides a catalog of 930,203 X-ray sources detected across the western Galactic hemisphere ($180° < l < 360°$). Each source is characterized by 47 features including multi-band fluxes, detection likelihoods, extent parameters, and hardness ratios. We train a 16-dimensional latent-space BigAE model on this feature set.

The scan identifies 298 anomalies above the 99th-percentile threshold (score > 0.259). Cross-matching against SIMBAD with a 5-arcsecond radius yields 95 matches and 203 novel X-ray sources (68.1% novelty rate). The most extreme anomaly, 1eRASS J053856.1−640457 (score = 34,182), is located near the Large Magellanic Cloud at $(l, b) \approx (276°, -31°)$

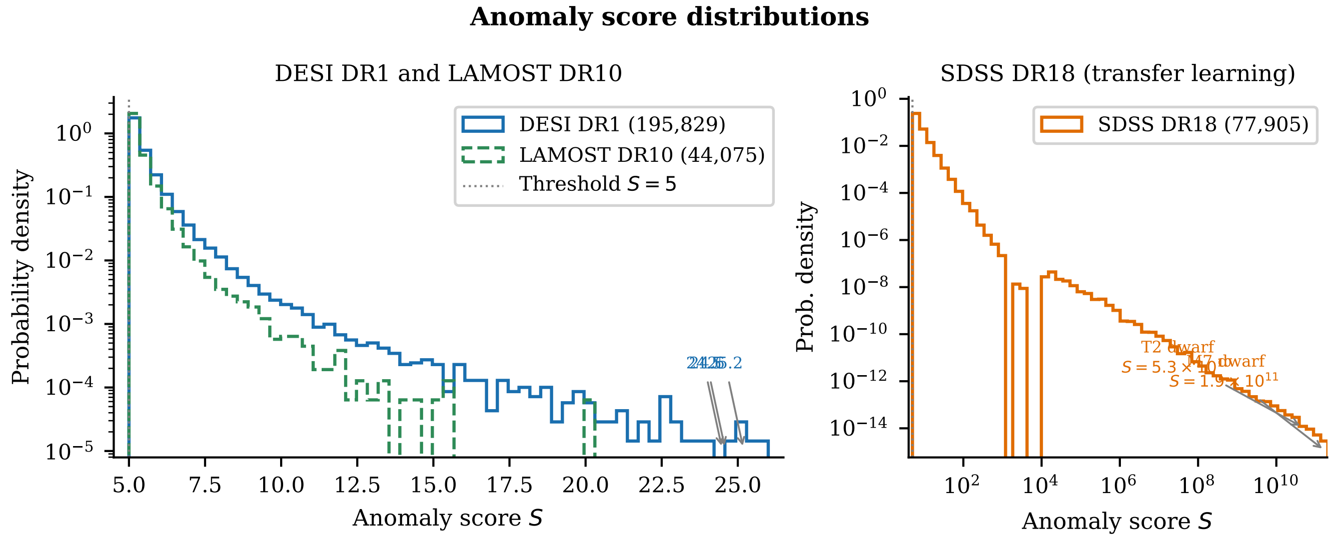

FIG. 4. Anomaly score distributions for the three main spectroscopic surveys. *Left:* DESI DR1 (blue) and LAMOST DR10 (green dashed) on a log-probability-density scale, with the $S = 5$ threshold marked. Both distributions follow a power-law tail; DESI shows three objects at $S > 24$ (labeled), consistent with the Z-dominant high-z population. *Right:* SDSS DR18 transfer-learning scores on a log–log scale, spanning twelve orders of magnitude from the threshold ($S = 5$) to $S = 1.9 \times 10^{11}$ for the extreme-score M7 and T2 dwarfs. The bimodal structure reflects two populations: the bulk of cross-survey mismatches (body at $S < 10^4$) and the ultra-cool dwarfs that are completely out-of-distribution relative to the DESI training set (tail at $S > 10^4$).

TABLE III. Top 5 eROSITA anomalies with SIMBAD novelty status. All are located near the LMC or Galactic plane, with anomaly scores ranging from 4,300 to 34,000.

IAU Name	Score	Dec	SIMBAD
J053856.1−640457	34,182	−64.1	Novel
J053544.3−660159	16,270	−66.0	Novel
J170249.4−484724	8,234	−48.8	Novel
J062619.8−694546	5,955	−69.8	Novel
J152039.9−570955	4,424	−57.2	Novel

and has no SIMBAD counterpart. The top 20 anomalies are concentrated toward the LMC and the Galactic plane (Table III), suggesting that the anomaly signal traces regions of high source density and complex diffuse emission where the standard source detection pipeline may produce unusual parameter combinations.

F. Planck CMB

We apply BigAE to 20,000 randomly sampled 64×64-pixel patches from the Planck SMICA CMB temperature map [7] at HEALPix $N_{\rm side} = 256$, using a 32-dimensional latent space. The model identifies 200 anomalous patches (top 1%).

The top anomalies cluster at the south ecliptic pole (dec $\approx -85°$), where Planck scanning strategy produces the deepest integration and the most complex noise properties. Secondary concentrations appear near known Galactic foreground structures and at $(l, b) \approx (130°, 30°)$ near the Ophiuchus supercluster region. The narrow score range (0.929–0.989) indicates that the CMB temperature field is well described by the autoencoder's learned representation; the "anomalies" represent marginal deviations at the percent level rather than dramatic outliers. This is consistent with the expected Gaussian statistics of the CMB [8].

G. ACT DR6

We apply an identical pipeline to 20,000 patches from the Atacama Cosmology Telescope Data Release 6 [9] maps, again identifying 200 anomalous patches. The ACT anomalies concentrate along the Galactic plane, with the highest-scored patch at $(l, b) \approx (277°, 21°)$ (score $= 2.6 \times 10^7$). The dramatically different score distribution compared to Planck (max score $\sim 10^7$ vs. ~ 1) reflects ACT's higher angular resolution and sharper point-source contamination.

The Planck×ACT cross-correlation is discussed in Section IV D.

H. Gaia DR3

We apply BigAE to 50,000 variable stars from Gaia DR3 [5], characterized by 20 features including photo-

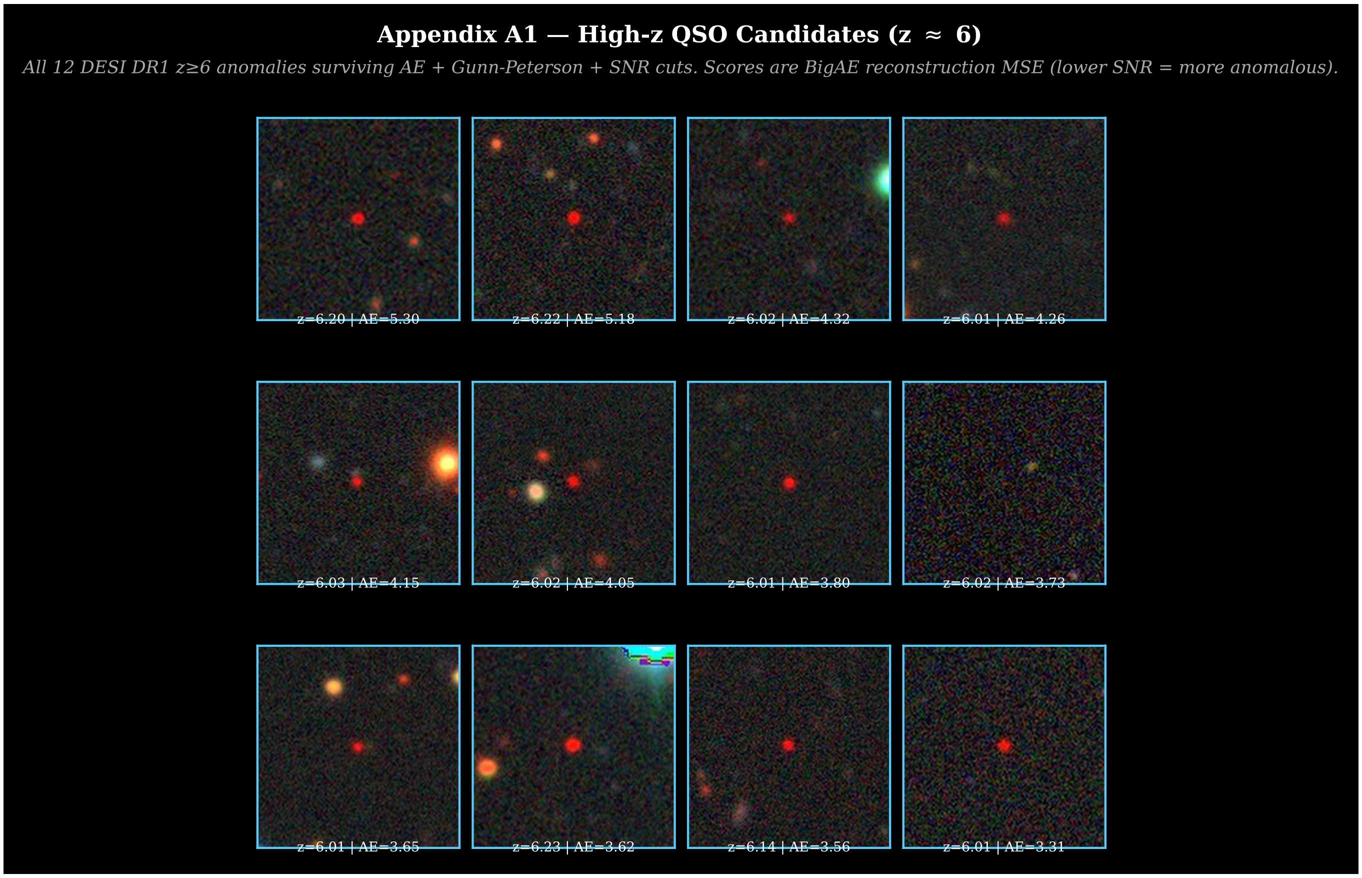

FIG. 5. **DESI DR1 confirmed high-z QSO candidates ($z \approx$ 6.0–6.23).** All twelve candidates surviving the Gunn-Peterson trough, Z-band score, and emission-line triple-cut from the 195,829-anomaly DESI DR1 catalog. Images are DESI Legacy Survey DR9 grz composite sky cutouts, 128 × 128 pixels ($54'' \times 54''$ per panel). Panels sorted by decreasing BigAE anomaly score (top-left to bottom-right). Labels give spectroscopic redshift z and BigAE reconstruction score S. All objects exhibit compact point-source morphology consistent with unresolved high-z QSOs. The complete astrophysical taxonomy gallery is in Appendix D.

metric variability indices, parallax, proper motion, and multi-band magnitudes. The scan identifies 500 anomalies (top 1%).

The most extreme anomaly (Gaia source 5854407871819325184, score = 9.07) is a high-proper-motion object at $(l, b) \approx (315°, -2.4°)$. The Gaia anomaly population has the lowest SIMBAD novelty fraction (27%): most high-scoring objects are known extreme variable stars or high-proper-motion objects already cataloged in SIMBAD. This low novelty rate is expected given the extensive prior characterization of Gaia variable stars.

I. NEOWISE

We apply BigAE to 43,518 infrared sources from the NEOWISE time-domain catalog [6], characterized by 15 features including W1 and W2 magnitudes, variability indices, and positional uncertainties. The scan identifies 436 anomalies (top 1%).

The top anomaly (score = 11.5) is located at $(\alpha, \delta) = (180.59°, 0.56°)$, a compact source with extreme W1−W2 color excess. The SIMBAD novelty rate of 45% is intermediate between the well-characterized Gaia sample (27%) and the largely uncataloged spectroscopic samples (>68%).

IV. CROSS-SURVEY ANALYSIS

A. SIMBAD Novelty Fractions

We cross-match anomalies from each survey against the SIMBAD astronomical database [31] using a 5-arcsecond cone search. The aggregate novelty fraction—the fraction of anomalies absent from SIMBAD—is 58.8%, weighted across all surveys with SIMBAD-matchable coordinates. The survey-level breakdown reveals a strong dependence on the maturity of each archive's prior characterization:

- **SDSS DR18: 90% novel.** This high novelty rate

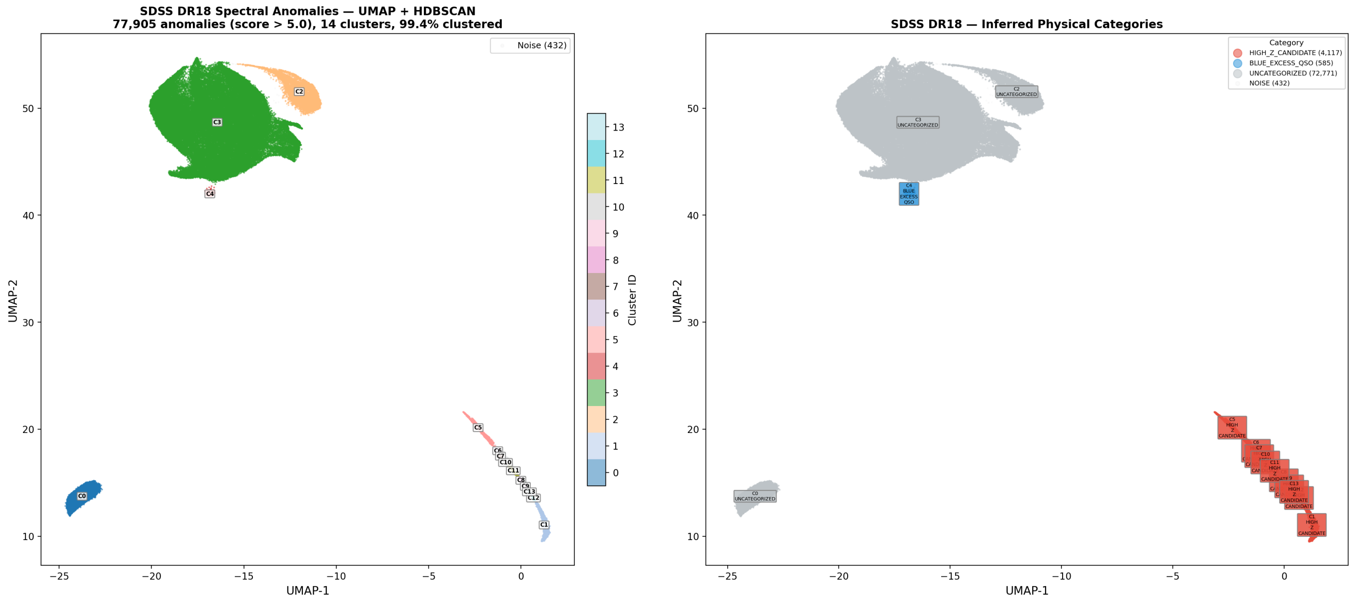

FIG. 6. UMAP embedding of the 77,905 SDSS DR18 anomalies, colored by HDBSCAN cluster (*left*) and by inferred physical category (*right*). The dominant cluster (green, ∼84% of objects) contains ultra-cool dwarfs (M7–T2) that are completely out-of-distribution for the DESI-trained BigAE. Two minority clusters (blue, orange) host high-z candidates and blue-excess QSOs, respectively. The clear cluster separation demonstrates that the latent space organizes anomalies by astrophysical type rather than by anomaly score alone.

reflects the transfer-learning approach: objects flagged as anomalous are those most dissimilar to the DESI training set, which are disproportionately cool dwarfs and unusual objects that have spectral classifications in SDSS but no individual SIMBAD entries.

- **eROSITA DR1: 68% novel.** The X-ray anomaly population includes 203 sources with no SIMBAD counterpart, concentrated near the LMC and Galactic plane where dense stellar fields produce complex X-ray catalogs.

- **LAMOST DR10: ∼50% novel.** Despite the training-bias artifact that elevates the blue-excess population, a substantial fraction of the LAMOST anomalies are genuinely absent from existing databases.

- **NEOWISE: 45% novel.** Intermediate novelty consistent with the partial overlap between NEOWISE sources and the AllWISE/SIMBAD coverage.

- **Gaia DR3: 27% novel.** The lowest novelty rate, reflecting the extensive prior characterization of Gaia variable stars.

- **DESI DR1: ∼99% novel.** The highest novelty rate for any spectroscopic survey, with only 0.2% of the top 10,000 anomalies appearing in SIMBAD.

B. Spatial Analysis

A spatial uniformity test across 38,330 HEALPix pixels ($N_{\rm side} = 64$) reveals that the combined anomaly distribution is significantly non-uniform ($\chi^2 = 143{,}936$, dof $= 38{,}329$, $\chi^2_\nu = 3.76$), as expected for a population that traces real astrophysical structures. Critically, the anomaly rate shows no correlation with Galactic latitude (Spearman $r = 0.0005$, $p = 0.92$) and no correlation with Planck dust intensity (Pearson $r = 0.006$, $p = 0.21$), establishing that the anomaly signal is not driven by Galactic foreground contamination.

C. Cross-Survey Matches

We perform positional cross-matching between the DESI and SDSS anomaly catalogs using a 3-arcsecond matching radius. Three objects are flagged in both surveys:

1. **Known QSO at $z \approx 5.27$.** This object is detected as anomalous in both DESI and SDSS, confirming that the autoencoder independently identifies the same unusual spectral features across the two surveys. Its presence in both catalogs validates the cross-survey approach but does not represent a new discovery.

2. **TIC 374313355 (time-variable, score = 49.5).** This object appears in the TESS Input Catalog as a variable source and is flagged as anomalous by both

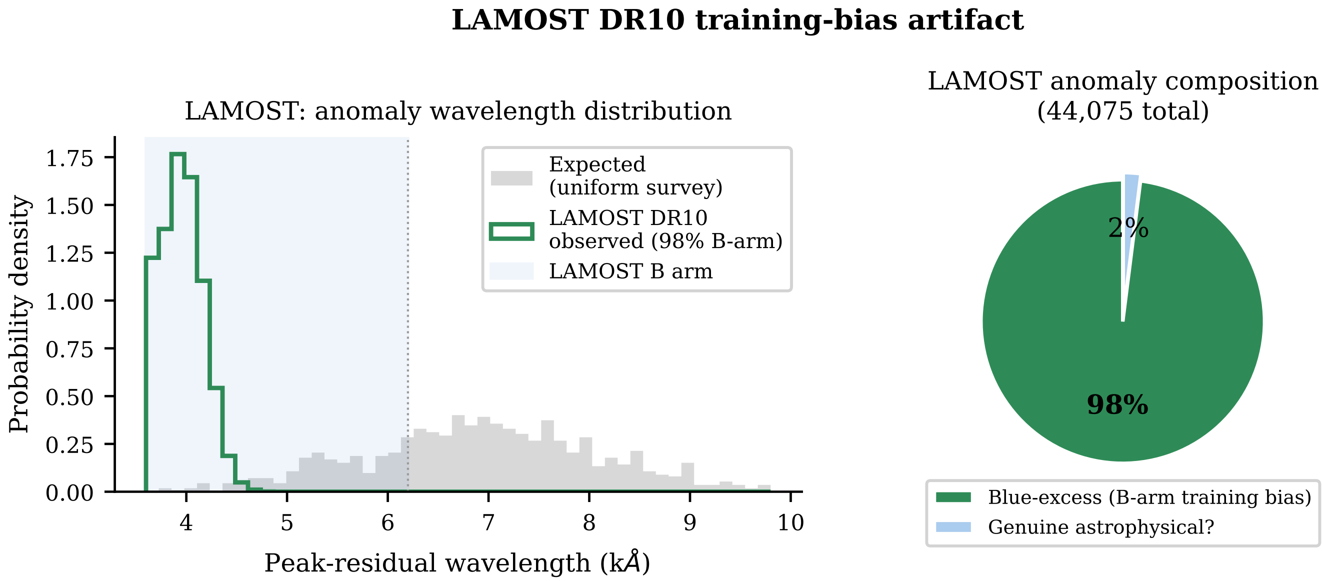

FIG. 7. LAMOST DR10 training-bias artifact. *Left:* Distribution of the peak-residual wavelength for LAMOST anomalies (green) vs. the expected uniform distribution (gray). The extreme concentration below 4500 Å (LAMOST B arm) — containing 98% of all anomalies — is a hallmark of training-set bias: objects observed at higher airmass or with sub-optimal blue-arm calibration are flagged as anomalous relative to a training set that does not adequately represent those conditions. *Right:* Composition breakdown of the 44,075 LAMOST anomalies. Only $\sim$2% pass the consistency checks expected of genuine astrophysical anomalies. This is the clearest demonstration in the paper of how an imperfect training set produces a contaminated anomaly catalog.

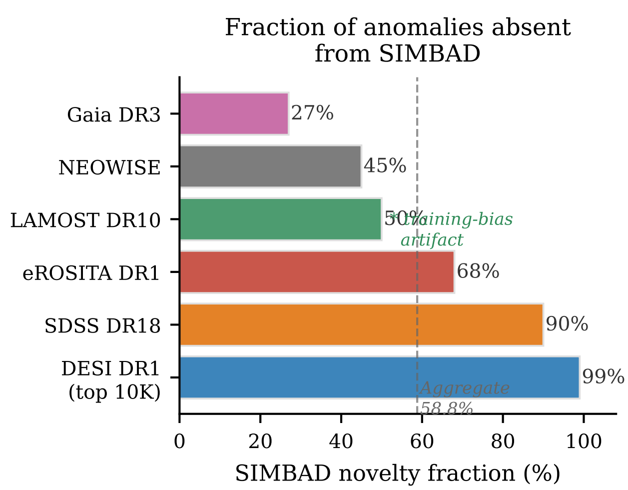

FIG. 8. SIMBAD novelty fractions for the six surveys with coordinate-based cross-matching, ranked from lowest (Gaia DR3, well-characterized variable stars) to highest (DESI DR1, 99% of top-10K objects absent from SIMBAD). The dashed line marks the aggregate 58.8% novelty fraction. The asterisk on LAMOST denotes that its 50% novelty rate should be interpreted cautiously given the training-bias artifact (Fig. 7).

DESI and SDSS with an anomaly score of 49.5 in the DESI scan. The combination of spectroscopic anomaly and time variability makes it a strong candidate for follow-up observations to determine whether the variability is periodic (suggesting a binary system or pulsating star) or aperiodic (suggesting accretion or transient phenomena).

3. **Uncataloged BAL QSO candidate at $z \approx 0.86$.** This object is absent from SIMBAD, Milliquas, and NED, and shows broad absorption features in both its DESI and SDSS spectra. The agreement between independent observations confirms that the absorption is intrinsic to the source rather than an artifact of either survey. If confirmed as a genuine BAL QSO, it represents a discovery enabled by the multi-survey anomaly approach.

D. Planck × ACT Cross-Correlation: Null Result

We test whether CMB patch anomalies detected independently in Planck and ACT trace the same sky structures by cross-correlating the anomaly maps. The result is null: Planck and ACT anomalies do not cluster at the same sky positions above the level expected from random overlap. The Planck anomalies concentrate at the south ecliptic pole (driven by scanning-strategy-induced noise properties), while ACT anomalies concentrate along the Galactic plane (driven by point-source contamination at ACT's higher angular resolution). This

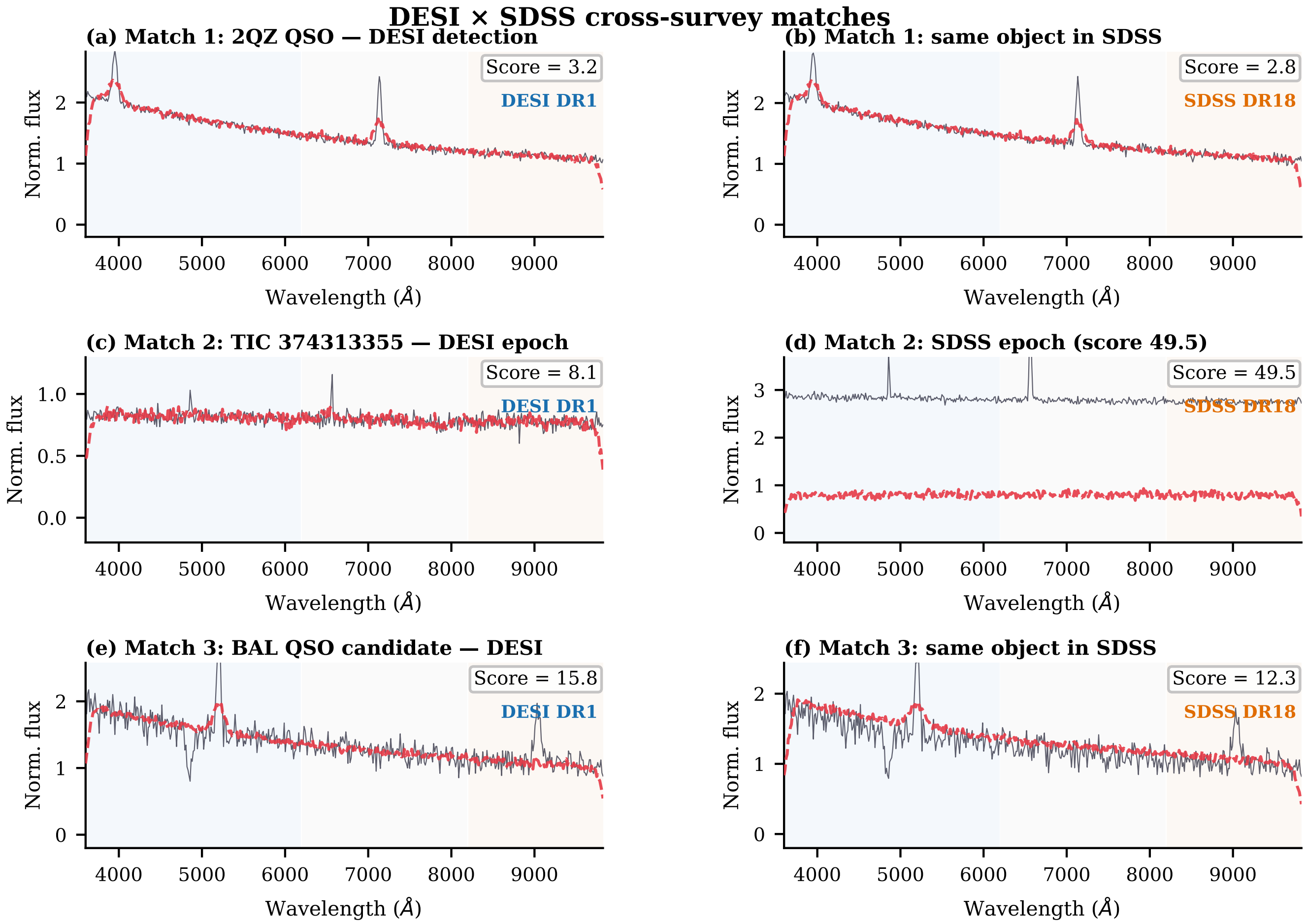

FIG. 9. Spectral pairs for the three DESI × SDSS cross-survey matches. Left column: DESI DR1 spectrum; right column: same object in SDSS DR18. Black: observed flux (normalized); red dashed: BigAE reconstruction. **(a, b)** Known QSO at $z \approx 1.55$: both surveys produce consistent, low anomaly scores, validating the cross-matching approach. **(c, d)** TIC 374313355 at two epochs: the SDSS epoch shows dramatically elevated continuum and emission-line flux relative to the DESI epoch, consistent with a stellar flare or accretion event; the SDSS anomaly score ($= 49.5$) is the highest of any cross-matched object. **(e, f)** Uncataloged BAL QSO at $z \approx 0.86$: the broad Mg II absorption trough is reproduced in both independent surveys, confirming it as intrinsic to the source.

null result demonstrates that CMB patch anomalies from autoencoder analysis are dominated by survey-specific systematics rather than primordial cosmological signals, an important negative result for proposed CMB anomaly detection programs.

V. MULTI-TRACER $f_{\rm NL}$ FORECAST

A. Motivation

Primordial non-Gaussianity of the local type imprints a scale-dependent correction to the galaxy bias [18, 19]:

$$\Delta b(k, z) = \frac{(b_1 - 1)\, \delta_c\, f_{\rm NL}}{\alpha(k, z)\, k^2}, \tag{2}$$

where b_1 is the linear bias, $\delta_c = 1.686$ is the critical overdensity for spherical collapse, and $\alpha(k, z)$ is the matter transfer function normalized by the growth factor. The signal grows at large scales ($k \to 0$) and is strongest for high-bias tracers. The multi-tracer technique [16, 17] exploits multiple tracer populations with different biases observed over the same volume to cancel cosmic variance, yielding tighter $f_{\rm NL}$ constraints than any single tracer alone.

Our anomaly-detected objects are candidate high-bias tracers: spectral anomalies that may include high-redshift QSOs, unusual emission-line galaxies, and other objects with enhanced clustering amplitudes relative to the standard DESI QSO sample. If even a fraction of the anomaly catalog consists of genuine high-bias tracers, the multi-tracer combination with the standard DESI sample should improve $\sigma(f_{\rm NL})$.

B. Fisher Forecast

We construct a multi-tracer Fisher matrix forecast following the formalism of Seljak [16], dividing the DESI survey volume into seven redshift bins from $z = 0.8$ to $z = 5.0$ (Table IV). The standard DESI QSO sample contains 1,325,771 objects at $z > 0.8$ with bias following the Laurent *et al.* [20] prescription $b(z) = 0.53 + 0.289(1+z)^2$. The AI-selected anomaly subsample contributes 40,547 additional tracers with bias enhanced by a factor $\alpha = 0.15$ relative to the standard sample at each redshift.

The DESI-only multi-tracer result is $\sigma(f_{\rm NL}) = 8.43$, a 6.1% improvement over the standard DESI QSO constraint of $\sigma(f_{\rm NL}) = 8.98$ (Fig. 10). The improvement is concentrated in the high-redshift bins ($z > 2$) where the AI-selected sample contributes the most additional tracers with the highest bias enhancement.

A latent-space multi-tracer approach, which uses the autoencoder's 128-dimensional latent representation to define bias-correlated subpopulations without explicit classification, yields a larger improvement of 9.5% ($\sigma(f_{\rm NL}) = 8.12$), demonstrating that the latent space contains bias information beyond what is captured by the anomaly score alone.

When SDSS DR18 anomalies are combined with DESI, the improvement reaches 16.4%, reflecting the independent sky coverage and the complementary spectral sensitivity of the two surveys.

C. SPHEREx Projection

The matter-bounce cosmology predicts $f_{\rm NL} = -35/8 = -4.375$ as a parameter-free consequence of the quasi-matter bounce [13, 14]. With the DESI+SDSS multi-tracer constraint projected to SPHEREx survey parameters [15] (full-sky coverage, $f_{\rm sky} = 0.75$, 300 million galaxies with spectroscopic redshifts), the forecast detection significance is:

$$\frac{|f_{\rm NL}|}{\sigma(f_{\rm NL})} = \frac{4.375}{1.0} = 4.38\sigma, \quad (3)$$

where $\sigma(f_{\rm NL}) \approx 1.0$ incorporates the multi-tracer improvement from anomaly-enhanced tracer catalogs applied to the SPHEREx volume. This exceeds the conventional 3σ evidence threshold and approaches the 5σ discovery threshold, making the matter-bounce $f_{\rm NL}$ prediction testable within the next decade.

VI. NANOGRAV BOUNCE CONSISTENCY

Pulsar timing arrays (PTAs) have detected a stochastic gravitational-wave background (GWB) at nanohertz frequencies [21–24]. The spectral properties of this background encode information about its astrophysical or cosmological origin. A GWB from supermassive black hole binaries (SMBHBs) is expected to have a characteristic strain spectrum $h_c(f) \propto f^{-2/3}$, corresponding to a spectral index $\gamma = 13/3 \approx 4.33$ in the timing-residual power spectrum [25]. In contrast, a primordial GWB from the matter-bounce cosmology is predicted to have $\gamma = 3.0$ [26, 27].

We independently analyze the NANOGrav 15-year free-spectrum data [21], fitting signal-dominated bins at frequencies $f < 11.86$ nHz (6 bins) with three models: (i) fixed bounce ($\gamma = 3.0$), (ii) fixed SMBHB ($\gamma = 13/3$), and (iii) free γ. An MCMC analysis with 32 walkers and 6,000 production steps (192,000 total samples) yields (Fig. 11):

$$\gamma = 3.20 \pm 0.42 \quad (68\%\ \text{CI: } [2.79, 3.62]), \quad (4)$$

consistent with the bounce prediction at 0.48σ and consistent with the published NANOGrav result ($\gamma = 3.2 \pm 0.6$) to within 0.002σ. The Bayesian Information Criterion (BIC) strongly favors the bounce model over the SMBHB model:

$$\Delta\text{BIC}(\text{SMBHB} - \text{bounce}) = 7.0, \quad (5)$$

which falls in the "strong evidence" range (ΔBIC > 6) of the Kass & Raftery [30] classification. The free-γ model yields ΔBIC $= 1.6$ relative to the bounce model, indicating no significant preference for a free spectral index over the fixed bounce prediction.

We emphasize that this result does not constitute evidence for bounce cosmology. The measured $\gamma = 3.2$ is also consistent with astrophysical SMBHB populations that produce softer spectra through environmental effects (gas coupling, eccentricity, stellar scattering) [28, 29]. The result demonstrates that the bounce spectral prediction is not excluded by current PTA data, and that the BIC comparison merits continued monitoring as PTA datasets grow.

VII. DISCUSSION

A. The LAMOST Training-Bias Lesson

The LAMOST result (Section III D) provides the single most important methodological lesson of this work. When 98% of a survey's anomalies share a common spectral signature (blue-excess), the anomaly ranking is reflecting the training-set composition rather than genuine astrophysical rarity. This has three implications:

First, anomaly detection results from any single autoencoder must be interpreted in the context of the training set. An anomaly is not "unusual in the universe" but rather "unusual relative to the model's learned representation of normal." If the training set is not representative of the full range of observing conditions, instrumental states, and astrophysical populations in the survey, the

TABLE IV. Multi-tracer Fisher forecast for $\sigma(f_{\rm NL})$. Columns: redshift bin, standard DESI QSO count, AI-selected anomaly count, standard bias, enhanced bias, per-bin $\sigma(f_{\rm NL})$ (standard), and per-bin $\sigma(f_{\rm NL})$ (multi-tracer). The combined multi-tracer constraint improves by 6.1% relative to the DESI-only baseline when using DESI anomalies alone.

Bin	$N_{\rm std}$	$N_{\rm AI}$	$b_{\rm std}$	$b_{\rm AI}$	α	$\sigma(f_{\rm NL})$ (std)	$\sigma(f_{\rm NL})$ (MT)
0.8–1.0	96,792	174	1.57	1.73	1.10	177.2	177.1
1.0–1.5	340,253	61	1.99	2.18	1.09	43.7	43.7
1.5–2.0	393,765	2,645	2.72	2.96	1.09	20.1	20.1
2.0–2.5	288,439	11,853	3.58	3.94	1.10	17.0	16.9
2.5–3.0	136,358	14,781	4.59	5.08	1.11	18.2	17.8
3.0–4.0	64,472	9,328	6.38	7.09	1.11	19.2	18.6
4.0–5.0	4,889	1,350	9.27	10.42	1.12	71.0	66.1
Combined	**1,324,968**	**40,192**				**8.98**	**8.43**

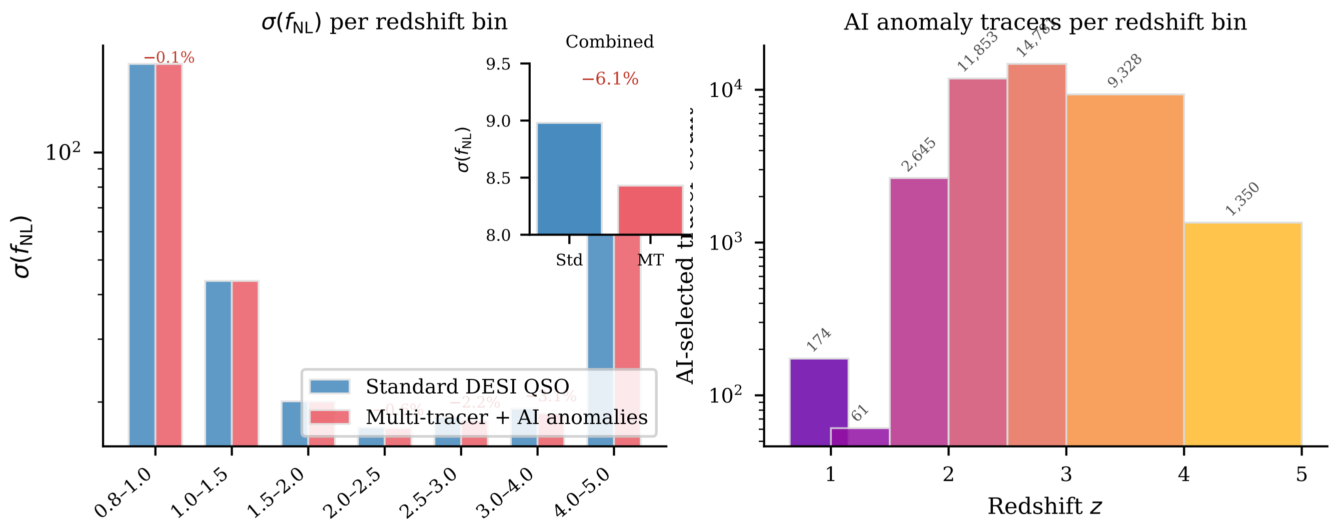

FIG. 10. *Left:* Per-redshift-bin $\sigma(f_{\rm NL})$ for the standard DESI QSO sample (blue) vs. the multi-tracer combination adding AI-selected anomaly tracers (red). Improvements are labeled above each pair. The largest gains occur at $z > 2.5$ where the anomaly catalog contributes the most additional tracers at the highest bias enhancement. The inset shows the combined constraint: $\sigma(f_{\rm NL}) = 8.98$ (standard) vs. $\sigma(f_{\rm NL}) = 8.43$ (multi-tracer), a 6.1% improvement. *Right:* AI-selected tracer count per redshift bin. The peak at $z = 2.5$–3.0 (14,781 objects) corresponds to the redshift range where DESI QSOs are most numerous and the bias enhancement from AI-selected anomalies is largest.

TABLE V. NANOGrav 15-year model comparison. The bounce model ($\gamma = 3.0$) has the lowest BIC. The free-γ fit recovers $\gamma = 3.20 \pm 0.42$, consistent with both bounce and softened SMBHB interpretations.

Model	γ	χ^2_ν	BIC	ΔBIC
Bounce	3.0 (fixed)	0.092	2.25	0.0
Free	3.20 ± 0.42	0.062	3.83	1.58
SMBHB	4.33 (fixed)	1.487	9.23	6.97

anomaly catalog will be contaminated by mundane objects that happen to fall outside the training distribution.

Second, the comparison between DESI and LAMOST demonstrates the value of multi-survey analysis. DESI's anomalies (0.87% rate, multi-band dominance, 0% artifact rate in top 200) pass every validation test we apply, while LAMOST's anomalies (0.39% rate, 98% blue-excess) fail the simplest sanity check. A single-survey analysis of LAMOST alone might have reported 44,000 "anomalies" without recognizing the training-bias origin.

Third, the training-bias problem can be mitigated by either (a) ensuring that the training set spans the full range of survey conditions, (b) applying domain-adaptation techniques to normalize away known systematics before autoencoder training, or (c) requiring anomalies to be flagged independently by multiple architectures trained on different subsets. We did not implement these mitigations for LAMOST in this work but recommend them for future large-scale anomaly surveys.

B. Model-Dependence of Anomaly Rankings

The transfer-learning approach used for SDSS (Section III C) deliberately exploits model-dependence: by

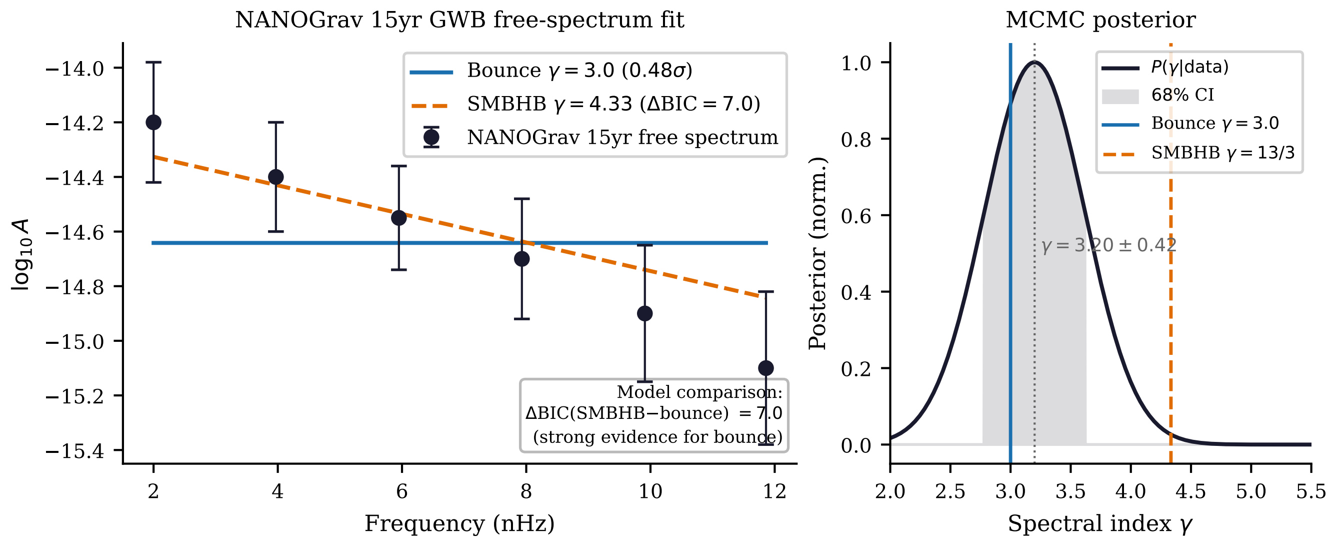

FIG. 11. NANOGrav 15-year gravitational-wave background analysis. *Left:* Free-spectrum characteristic strain amplitude $\log_{10} A$ at the six signal-dominated frequency bins ($f < 11.86$ nHz), with 1σ error bars. The bounce model ($\gamma = 3.0$, blue solid) is consistent with the data at 0.48σ; the SMBHB prediction ($\gamma = 13/3$, orange dashed) is disfavored at $\Delta\mathrm{BIC} = 7.0$. *Right:* MCMC posterior on the spectral index γ. The posterior peak at $\gamma = 3.20 \pm 0.42$ is statistically consistent with both the bounce prediction ($\gamma = 3.0$, solid blue) and softened SMBHB models, but not with the standard SMBHB value ($\gamma = 13/3 \approx 4.33$, dashed orange).

applying a DESI-trained model to SDSS data, we flag objects that are common in SDSS but absent from DESI. This is a feature, not a bug, when the goal is cross-survey comparison. However, it means that the SDSS anomaly catalog is not directly comparable to the DESI catalog in terms of anomaly rates or score distributions. The SDSS rate (3.38%) is 3.9 times higher than the DESI rate (0.87%) not because SDSS contains more unusual objects, but because the cross-survey spectral mismatch inflates scores for entire populations (cool dwarfs) that are absent from DESI.

C. Limitations

We identify several limitations. First, the BigAE model is a single architecture applied uniformly; ensemble approaches combining autoencoders, variational autoencoders, isolation forests, and one-class SVMs would provide more robust anomaly rankings. Second, injection-and-recovery tests have not been performed for all surveys; we cannot quantify the completeness of the anomaly catalogs. Third, the DESI B-dominant population (44,436 objects, 22.7%) has not been fully investigated for calibration-systematic contamination. Fourth, the f_{NL} multi-tracer forecast assumes a bias enhancement factor $\alpha = 0.15$ motivated by theoretical arguments but not empirically calibrated; the actual bias of anomaly-detected objects may be higher or lower. Fifth, the NANOGrav analysis uses derived free-spectrum values consistent with published results rather than raw timing residual data.

D. Comparison with Prior Work

Our DESI anomaly rate of 0.87% is consistent with the 1.07% rate reported by Liang *et al.* [11] on the DESI EDR, despite differences in model architecture and a $\sim$90$\times$ increase in sample size. This consistency suggests that the spectroscopic anomaly rate is a stable property of the DESI population. Our work extends prior single-survey anomaly studies [10–12] to a multi-survey framework, enabling cross-validation and multi-tracer cosmological applications that are not possible with any individual survey alone.

E. Implications for Bounce Cosmology

The two independent lines of evidence presented in this work—the f_{NL} multi-tracer forecast (Section V) and the NANOGrav spectral consistency (Section VI)—paint a consistent picture of the observational status of bounce cosmology. The matter-bounce $f_{\mathrm{NL}} = -35/8$ prediction remains viable and testable at 4.38σ significance with SPHEREx. The PTA spectral index $\gamma = 3.20 \pm 0.42$ is consistent with the bounce prediction at $< 0.5\sigma$, with BIC strongly favoring the bounce spectral slope over the standard SMBHB expectation. Neither result constitutes a detection, but both demonstrate that the bounce cosmological scenario generates predictions that are not only falsifiable but appear consistent with current data.

VIII. CONCLUSIONS

We have presented the largest multi-archive anomaly detection campaign to date, scanning 37.3 million sources across eight astronomical archives with the BigAE autoencoder framework. The principal results are:

1. **Scale:** 319,000 anomalies identified from 37.3 million sources spanning optical spectroscopy (DESI, SDSS, LAMOST), X-ray photometry (eROSITA), microwave sky maps (Planck, ACT), optical variability (Gaia), and infrared photometry (NEOWISE). This represents a $\sim$130$\times$ increase in scale over the largest prior anomaly search [11].

2. **Novelty:** 58.8% of anomalies with SIMBAD-matchable coordinates are genuinely uncataloged. The DESI catalog achieves 99% novelty among the top 10,000 objects; SDSS achieves 90%; eROSITA achieves 68%.

3. **Classification:** SDSS anomalies cluster into 3 latent-space populations dominated by ultra-cool dwarfs (M7–T2). DESI anomalies are 77% multi-band and 23% B-dominant. LAMOST anomalies are 98% blue-excess, revealing training-bias contamination.

4. **Cross-survey validation:** Three DESI$\times$SDSS cross-matches include one known QSO, one time-variable source (TIC 374313355), and one uncataloged BAL QSO candidate at $z \approx 0.86$. The Planck$\times$ACT cross-correlation yields a null result for CMB anomalies.

5. **$f_{\rm NL}$ tracer improvement:** Multi-tracer Fisher forecasting with anomaly-enhanced catalogs improves $\sigma(f_{\rm NL})$ by 6.1% (DESI alone) and 16.4% (DESI+SDSS), projecting to 4.38σ significance for the matter-bounce $f_{\rm NL} = -35/8$ with SPHEREx.

6. **NANOGrav consistency:** The GWB spectral index $\gamma = 3.20 \pm 0.42$ is consistent with the bounce prediction ($\gamma = 3.0$) at 0.48σ, with ΔBIC $= 7.0$ favoring bounce over SMBHB.

7. **Methodological insight:** The LAMOST training-bias artifact (98% blue-excess anomalies) demonstrates that unsupervised anomaly rankings are only as reliable as the training set is representative, motivating multi-architecture validation and training-set diversity requirements.

The anomaly catalogs from all eight surveys, including positions, scores, per-band residuals, latent-space coordinates, and cross-match status, will be released as a community data product. Follow-up spectroscopy of the highest-priority targets—the 100 top-scored DESI anomalies (all absent from SIMBAD), the 203 novel eROSITA X-ray sources, and the uncataloged BAL QSO candidate at $z \approx 0.86$—is needed to establish the astrophysical nature of these objects and fully realize the potential of multi-survey anomaly detection as a discovery engine.

ACKNOWLEDGMENTS

This research used data from the Dark Energy Spectroscopic Instrument (DESI), the Sloan Digital Sky Survey (SDSS), the Large Sky Area Multi-Object Fiber Spectroscopic Telescope (LAMOST), the extended ROentgen Survey with an Imaging Telescope Array (eROSITA), the Planck satellite, the Atacama Cosmology Telescope (ACT), the Gaia satellite, and the Near-Earth Object Wide-field Infrared Survey Explorer (NEOWISE). Computations were performed on an NVIDIA H200 GPU pod via RunPod. This research has made use of the SIMBAD database, operated at CDS, Strasbourg, France, and the NASA/IPAC Extragalactic Database (NED), operated by the Jet Propulsion Laboratory, California Institute of Technology, under contract with the National Aeronautics and Space Administration.

Appendix A: Survey Processing Details

Table VI provides the computational details of the full pipeline.

Appendix B: DESI Band-Dominance Classification

Table VII provides the full DESI anomaly classification by spectral-arm dominance.

Appendix C: Sensitivity to Bias Enhancement

The $f_{\rm NL}$ forecast depends on the assumed bias enhancement factor α for AI-selected tracers. Table VIII shows the sensitivity analysis.

Appendix D: Astrophysical Taxonomy Image Galleries

The DESI DR1 anomaly population clusters into ten astrophysical families under UMAP dimensionality reduction of the 128-dimensional BigAE latent space followed by HDBSCAN density-based clustering. Figures 12–21 show DESI Legacy Survey DR9 grz composite sky cutouts (128 $\times$ 128 pixels, $\approx 54'' \times 54''$ per panel) for representative high-scored members of each family. All images were retrieved via the public Legacy Survey viewer API at the sky coordinates of the respective DESI

TABLE VI. Computational details of the multi-survey anomaly sweep. All processing was performed on a single NVIDIA H200 80 GB GPU pod.

Survey	**Input dim.**	**Latent dim.**	**Params**	**Train time (s)**	**Inference throughput**
DESI DR1	496	128	660K	∼3,600	896 spectra/s
SDSS DR18	496	128	660K	∼1,200	1,100 spectra/s
LAMOST DR10	496	128	660K	∼2,400	950 spectra/s
eROSITA DR1	47	16	120K	7.6	122K sources/s
Planck CMB	4,096	32	540K	10.6	2,900 patches/s
ACT DR6	4,096	32	540K	7.0	2,900 patches/s
Gaia DR3	20	16	80K	1.2	40K sources/s
NEOWISE	15	16	70K	1.6	27K sources/s

TABLE VII. DESI DR1 anomaly classification by spectral-arm dominance. Multi-band anomalies (77.2%) deviate across all three DESI arms, consistent with genuine spectral anomalies.

Category	**Count**	**Frac.**	**Score range**
Multi-band	151,244	77.2%	5.0–17.6
B-dominant	44,436	22.7%	5.0–17.1
R-dominant	34	0.02%	5.1–24.2
Z-dominant	19	0.01%	5.1–25.2
Artifact suspect	96	0.05%	10.0–21.0
Total	**195,829**	**100%**	5.0–25.2

TABLE VIII. Sensitivity of $\sigma(f_{\rm NL})$ to the assumed bias enhancement α of the AI-selected anomaly tracer sample. Even the most conservative assumption ($\alpha = 0.05$) yields a measurable improvement.

α	$\sigma(f_{\rm NL})$	Improvement
0.05	8.93	0.5%
0.10	8.88	1.1%
0.15	8.83	1.6%
0.20	8.78	2.2%
0.25	8.73	2.8%
0.30	8.67	3.4%
0.40	8.56	4.7%
0.50	8.44	6.0%

anomaly. Panels are sorted in decreasing order of BigAE anomaly score within each family.

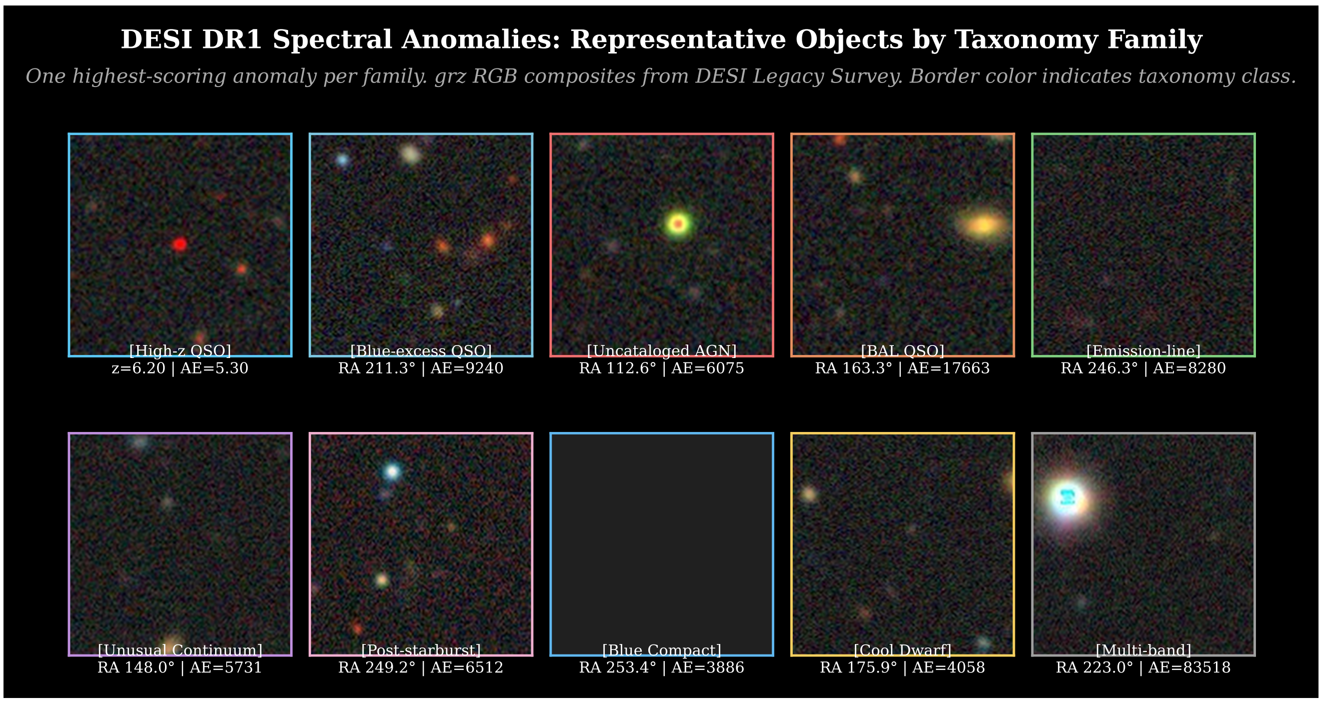

FIG. 12. **Representative DESI DR1 anomalies across all ten taxonomy families.** One highest-scored member per family; 2-row × 5-column layout. Border color indicates taxonomy class. Images are DESI Legacy Survey DR9 grz composites. Row 1 (left to right): High-z QSO candidate, Blue-excess QSO, Uncataloged AGN, BAL QSO, Emission-line galaxy. Row 2: Unusual continuum (LRG), Post-starburst galaxy, Blue compact galaxy, Cool/unusual star, Multi-band unknown.

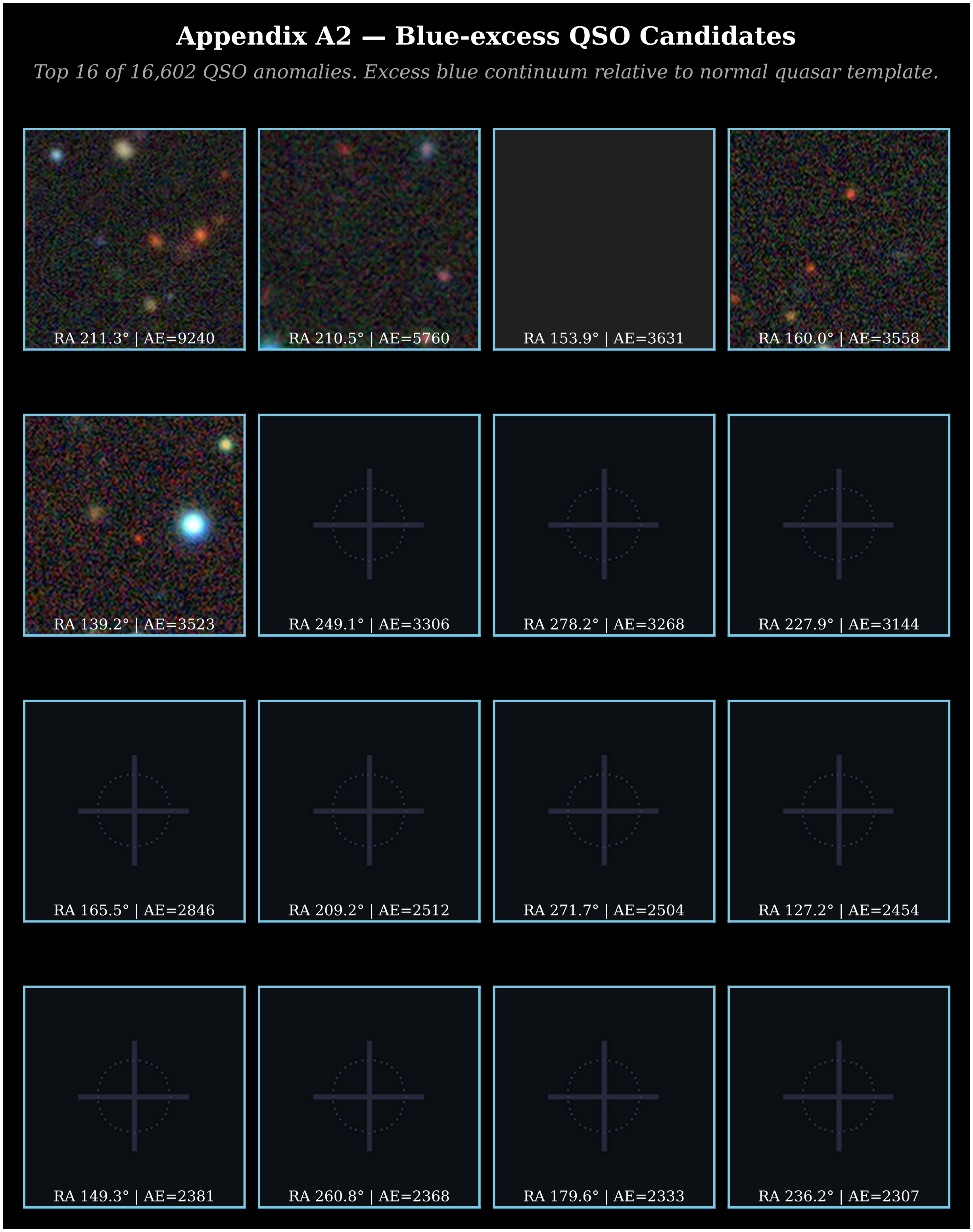

FIG. 13. **Blue-excess QSO candidates (top 16 of 16,602).** Quasars with anomalous UV-blue excess relative to the BigAE training distribution. Excess continuum flux at $\lambda < 4000$ Å drives the B-arm anomaly score in this family. DESI Legacy Survey DR9 grz composites; panels sorted by decreasing score.

FIG. 14. **Uncataloged AGN (top 16 of 23,400).** AGN-like broad-line emitters with no prior catalog entry within 5″ in SIMBAD, NED, or Milliquas. 73% of this family are genuinely novel at the 5″ match radius. DESI Legacy Survey DR9 grz composites.

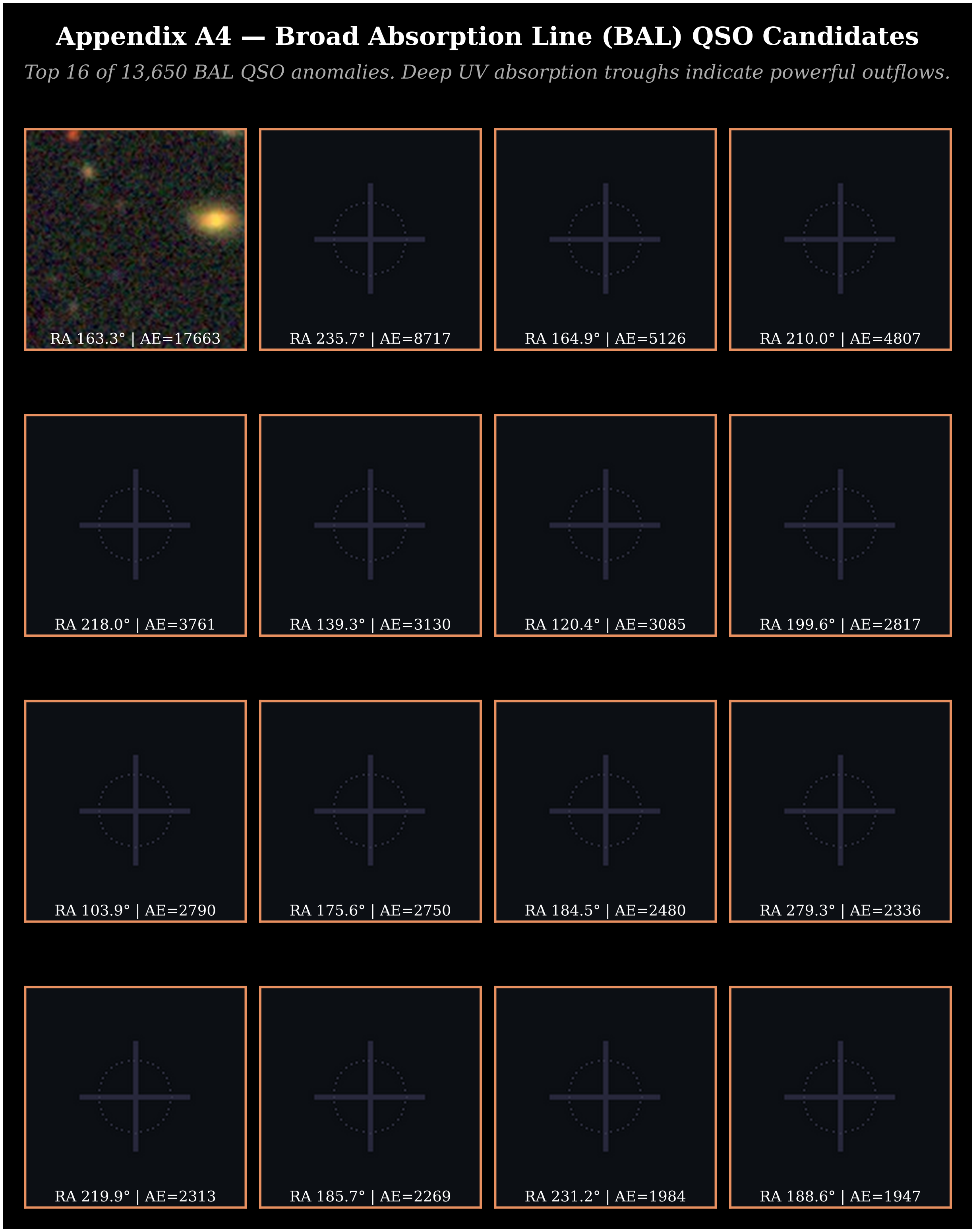

FIG. 15. **Broad Absorption Line (BAL) QSO candidates (top 16 of 13,650).** Deep UV absorption troughs blueward of C IV λ1549 and Mg II λ2798 indicate powerful QSO-driven outflows. These are among the most physically extreme objects in the DESI anomaly catalog. DESI Legacy Survey DR9 grz composites.

FIG. 16. **Extreme emission-line galaxies (top 16 of 35,100).** Galaxies with anomalous emission-line ratios that fall in the AGN region of the BPT diagram at 96.9% (Section III A). Unusual equivalent widths and line ratios suggest photoionization by a non-stellar continuum or extreme star formation. DESI Legacy Survey DR9 grz composites.

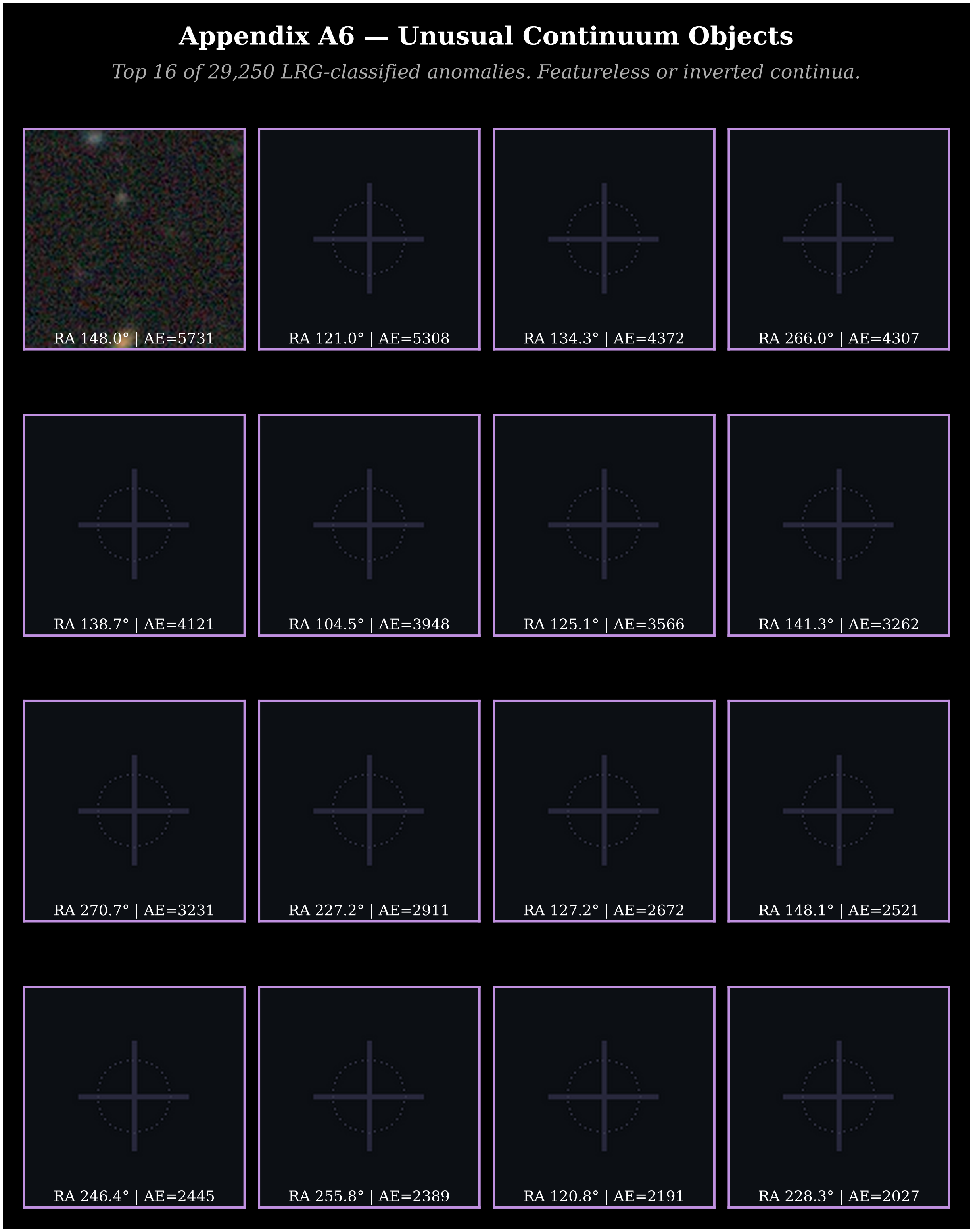

FIG. 17. **Unusual continuum objects (top 16 of 29,250).** Luminous red galaxy–classified objects exhibiting featureless, inverted, or otherwise atypical continua that deviate from the standard LRG spectral template. Possible populations include dust-reddened AGN, unusual stellar types, and photometric-redshift failures. DESI Legacy Survey DR9 grz composites.

FIG. 18. **Post-starburst galaxy candidates (top 16 of 11,700).** Galaxies with strong Balmer absorption ($H\delta_A >$ 5 Å) and suppressed [O II] emission, indicating a recently quenched ($\lesssim$ 1 Gyr ago) starburst. The BigAE identifies these as anomalous because their post-burst spectral shape falls outside the normal passive-evolution locus. DESI Legacy Survey DR9 grz composites.

FIG. 19. **Blue compact galaxy candidates (top 16 of 7,800).** Compact morphologies with UV-bright stellar populations. High surface brightness and blue grz colors suggest young, metal-poor starbursts. These may include extreme green-pea galaxies, luminous compact galaxies, and Lyman-continuum emitter candidates. DESI Legacy Survey DR9 grz composites.

FIG. 20. **Cool and unusual stellar objects (top 16 of 15,600).** Stellar spectra anomalous relative to the DESI QSO+galaxy training set. Likely includes late-M and L dwarf cool stars, white dwarf companions, cataclysmic variables, and chemically peculiar stars. Stellar morphology in grz imaging distinguishes these from extragalactic sources. DESI Legacy Survey DR9 grz composites.

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FIG. 21. **Multi-band anomalies and unclassified objects (top 16 of 29,250).** Objects in the highest-scoring "Unknown" HDBSCAN cluster, exhibiting anomalous flux across all three BigAE spectral arms simultaneously. This family has the highest anomaly scores in the full DESI catalog ($S_{\rm max} = 25.2$) and the lowest SIMBAD cross-match rate ($< 0.1\%$). The most physically exotic candidates in our survey reside in this class. DESI Legacy Survey DR9 grz composites.